

Intellectual Property Spatiotemporal Proof Protocol (IPSTP)

Engineering Implementation and Legal Interpretation Specification

Version: v3.0

Baseline Protocol: IPSTP v1.0

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Abstract

This document serves as the supporting engineering implementation and legal interpretation specification for the core IPSTP v1.0 protocol. Addressing the structural contradiction where "public disclosure easily leads to infringement, while confidentiality results in the loss of priority rights," IPSTP v1.0 established the core paradigm of "separation of verification rights and access rights." This v3.0 interpretive version aims to translate the minimalist rules of the core protocol into a standardized evidence structure that is implementable, machine-verifiable, and judicially admissible across jurisdictions. It clarifies the engineering boundaries of the three-layer disclosure, the technical specifications of spatiotemporal dual-

anchoring, and the evidentiary validity of the three-tier rights confirmation system in judicial burden of proof.

This document also provides a complete practical guide and toolchain appendix. Any researcher can independently complete the full-process implementation of the IPSTP protocol based on this document.

Keywords: Intellectual Property Spatiotemporal Proof Protocol; Condition–Conclusion Theoretical System; Three-Layered Disclosure; Three-Tier Gradient Rights Confirmation; Spatiotemporal Dual-Anchoring; Verifiable Disclosure Event; Scientific IP Evidence Preservation

1. Applicable Objects and Core Mapping

1.1 Condition–Conclusion Theoretical Systems (CCTS)

The core applicable objects of IPSTP are Condition–Conclusion Theoretical Systems (CCTS). The system consists of three elements:

- **Condition Set C:** Formalizable theoretical premises and parameter systems.
- **Constraint Set B:** Structural constraints, convergence conditions, and conservation rules of the system.
- **Conclusion Set Y:** Precise numerical conclusions and testable predictions derived from C and B.

The core characteristic of CCTS is that the dependence of conclusions on conditions is deterministic—different engineering implementation paths can reproduce consistent results within a reasonable margin of error. Therefore, the core intellectual

property value of such achievements is concentrated in the deterministic mapping relationship of $C \rightarrow Y$, rather than specific code, iterative algorithms, or engineering implementation details.

1.2 Mapping of Logical Structure and Disclosure Layers

To eliminate terminology confusion, this specification clarifies the mapping relationship between the content logic elements of CCTS and the IPSTP disclosure layers:

CCTS Element	IPSTP Layer	Disclosure Requirement
Condition Set C	Layer A (Condition & Conclusion Public Layer)	Mandatory Disclosure
Conclusion Set Y	Layer A (Condition & Conclusion Public Layer)	Mandatory Disclosure
Constraint Set B	Layer B (Constraint Framework Public Layer)	Mandatory Disclosure
Engineering Implementation	Layer C (Implementation Source Code Confidential Layer)	Default Confidentiality

This mapping relationship is a mandatory standard of IPSTP, and users shall not adjust it independently.

2. Engineering and Legal Interpretation of Core Rules

2.1 Definition of the Principle of Verification Sufficiency

The core protocol stipulates that "A+B must be sufficient to support independent reproduction of conclusions by third parties." In implementation and judicial review, this principle is defined as:

Based on the conditions disclosed in Layer A and the constraint framework disclosed in Layer B, a third party can, through an independent engineering implementation path, verify and obtain conclusion values and qualitative behaviors consistent with those of the original author.

Key legal and technical definitions:

- Verification does not require the third party to access or reuse the original author's Layer C source code.
- Layer C is absolutely not a necessary condition for verification.
- The technical choices of the third party's implementation path (programming language, numerical libraries, grid strategies, etc.) do not affect the validity of the verification conclusion.

2.2 Technical Specifications for Spatiotemporal Dual-Anchoring

Spatial Anchoring: Generating SHA-256 hash fingerprints for the public documents of Layers A/B and the encrypted package of Layer C to solidify the digital identity of the content, lock in version authenticity, and prevent tampering and replacement.

Operation Steps:

1. Save Layer A/B documents in PDF or Markdown format.
2. Package Layer C source code into an encrypted compressed file (AES-256 encryption recommended).
3. Calculate the SHA-256 hash values for the above files respectively.
4. Explicitly label the hash values in the verification layer lock information area of the public documents.

Temporal Anchoring: Relying on authoritative third-party Trusted Timestamps (TSA), blockchain evidence preservation, or official DOI registration to solidify the chronological sequence of the first public disclosure, forming an irreversible time coordinate of priority rights.

Operation Steps (in order of priority):

1. **TSA Timestamp:** Submit the SHA-256 hash value of the Layer A/B documents to a Trusted Timestamp Authority (TSA) compliant with the RFC 3161 standard (e.g., TSA.cn) to obtain a timestamp certificate.
2. **Blockchain Archiving:** Write the SHA-256 hash value into the OP_RETURN field or smart contract storage of public chains such as Ethereum, Tron, or BNB Chain, obtaining the transaction hash as evidence.
3. **DOI Registration:** Obtain a document DOI through platforms like Zenodo or arXiv, using the platform's official registration time as an auxiliary temporal anchor.

2.3 Verifiable Disclosure Event (VDE)

A VDE—the minimal public evidence unit of IPSTP—is generated if and only if the

following four rigid conditions are simultaneously met:

Condition	Verification Method
Complete theoretical and constraint systems of Layers A/B are public	Documents are publicly accessible
SHA-256 hash values of public files are fully retained and verified error-free	Can be independently calculated and compared
Authoritative third-party timestamp archiving takes effect, chronology is irreversible	TSA certificate or blockchain transaction is verifiable
Any third party can independently verify core conclusions based on public content	Does not require access to Layer C

The generation of a VDE objectively records a factual event of "disclosing conclusion Z with content Y at time X." The accumulation of multiple VDEs constitutes a complete priority evidence chain for the theoretical system.

2.4 Legal Closed-Loop of Retrieval and Confidentiality Mechanism

Layer C is permanently and privately preserved by default, fully applying Trade Secret protection. To prevent the confidentiality mechanism from being misinterpreted as "refusing judicial review," exceptions for retrieval exemption are clarified:

- **Judicial Evidence Submission:** Decrypted directionally for judicial review upon court investigation orders or evidence preservation rulings.
- **Administrative Investigation:** Cooperate in providing Layer C evidence upon investigation requirements from intellectual property administrative

protection departments.

- **Written Consent of the Right Holder:** The right holder may independently decide to disclose Layer C in scenarios such as arbitration or authorized audits.
- **Cooperative Research:** Directionally share Layer C content after signing a Non-Disclosure Agreement (NDA) with partners.

In the above statutory scenarios, Layer C is directionally decrypted and verified according to compliance procedures, serving as the ultimate fallback asset for intellectual property. Directional disclosure is not deemed as a waiver of the Trade Secret nature of Layer C.

3. Three-Tier Gradient Rights Confirmation System and Operating Specifications

Users select the rights confirmation level according to publication intensity and judicial admissibility requirements. The three-tier system is weighted progressively from light to heavy:

3.1 Tier 1: Routine Verification Level (Lightweight Academic Archiving)

Applicable Scenarios: Rapid first publication, preliminary priority archiving, academic announcement.

Operating Specifications:

1. Publish Layer A/B content through preprint platforms (arXiv, OSF Preprints, etc.) or open archiving platforms (Zenodo, Figshare, etc.) to obtain a DOI.
2. Save Layer C source code locally encrypted, without public disclosure.

3. Declare in the Layer A/B document: "This achievement complies with IPSTP v1.0 Tier 1 Confirmation."
4. Record and save the SHA-256 hash values of the Layer A/B documents.

Evidentiary Validity: Lightweight academic archiving level. Layer C can be retrieved as supplementary evidence during disputes, but the timestamp of a single preprint or archive has limited weight in judicial admissibility.

Upgrading to Tier 2 or 3 is recommended.

3.2 Tier 2: Evidence Anchoring Level (Standard Commercial-Grade, Officially Recommended)

Applicable Scenarios: Academic publication, project closure, routine IP rights protection, achievement registration.

Operating Specifications:

1. Host compliant Layer A/B content in a public GitHub repository.
2. Enable automatic archiving permissions for the repository on the Zenodo platform.
3. Publish a version via GitHub Release; the system automatically triggers: full repository snapshot capture, global SHA-256 hash generation, CERN multi-copy permanent archiving, multiple timestamp locking, and dual-DOI automatic issuance (Concept DOI + Version DOI).
4. Generate the SHA-256 hash value of the Layer C encrypted package and explicitly label it in the appendix of the published document.
5. Submit the SHA-256 of the Layer C encrypted package to TSA.cn for

timestamp solidification.

6. Declare in the Layer A/B document: "This achievement complies with IPSTP v1.0 Tier 2 Confirmation."
7. Label archiving DOI, TSA certificate number, and other evidence information.

Evidentiary Validity: Standard commercial-grade archiving level. Possesses global searchability, traceability, and permanent non-deletability. GitHub Release + Zenodo archiving + TSA timestamp constitutes a standard structured evidence chain.

Core Fallback Rule: Deletion or modification of the local GitHub repository will not alter or invalidate the Zenodo cloud permanent archive snapshot. Archive hashes, version chronology, and DOI records are permanently valid, possessing tamper-proof and irrevocable permanent evidentiary 效力 (validity).

3.3 Tier 3: Maximum Preservation Level (Highest Judicial Priority)

Applicable Scenarios: High-value core achievements, foreseeable priority disputes, cross-border IP evidence submission, judicial arbitration evidence submission.

Operating Specifications:

1. Streamline public disclosure of Layer A/B (sufficient for verification sufficiency, no need to disclose all details).
2. After completing spatial and temporal dual-anchoring, seal and encrypt Layer C.

3. Implement off-site multi-backup: at least three independent physical locations.
4. Superimpose multiple evidence preservation means (select at least three):
RFC 3161 standard TSA, Ant Chain blockchain archiving, ETH/TRX/BSC public chain anchoring, WIPO Proof, or Notary Office official electronic archiving.
5. Form a cross-validation matrix of encrypted package hashes, timestamp certificate numbers, and blockchain transaction hashes.
6. Mandatorily label the standard declaration in the main text of the Layer A/B document.

Mandatory Labeling Declaration Template:

"This paper complies with the IPSTP v1.0 Tier 3 Preservation Mechanism. The public content of Layers A/B has completed spatial and temporal anchoring. Archive Number: [Fill in]. TSA Timestamp Certificate No.: [Fill in]. Blockchain Transaction Hash: [Fill in]. The Layer C implementation assets are retained as Trade Secrets in accordance with IPSTP v1.0, and multi-layer encrypted evidence preservation has been completed."

Evidentiary Validity: Highest judicial admissibility weight. A strong evidence chain with multi-source cross-validation, meeting the evidence standards for judicial disputes, infringement burden of proof, and cross-border IP evidence submission.

4. Cross-Jurisdictional Legal Positioning and Evidence Adaptation

IPSTP does not create new legal rights or replace existing IP systems; it merely provides structured evidence chains to support existing judicial adjudication.

4.1 Common Law Adaptation (US, UK)

VDE public archiving records generated by IPSTP can serve as standard Prior Art evidence to oppose subsequent preemptive patent registrations. Layer C engineering implementation content fully adapts to the legal requirements of Trade Secret protection:

- **Secrecy:** Layer C is not public.
- **Protective Measures:** Encrypted archiving, access control.
- **Commercial Value:** Exclusivity of core deduction logic.

Meeting these three elements, Layer C content can be protected as a Trade Secret under Common Law for infringement liability pursuit.

4.2 EU Jurisdiction Adaptation

The overall architecture follows the technical requirements of eIDAS (Electronic Identification and Trust Services) for electronic evidence and trust services. Key definitions:

- When using qualified timestamps compliant with eIDAS issued by Qualified Trust Service Providers (QTSP), the relevant temporal anchoring records can be accepted as qualified electronic evidence within the EU.
- Archiving records like Zenodo and DOI can serve as supplementary evidence but do not themselves constitute eIDAS qualified timestamps.
- It is recommended to superimpose qualified timestamps issued by QTSP when used within the EU to enhance evidentiary validity.

4.3 Chinese Legal System Adaptation

The spatiotemporal anchoring and VDE mechanisms of IPSTP are fully compatible with China's current legal system:

- **Electronic Signature Law:** Trusted timestamps compliant with RFC 3161 possess legal validity.
- **Supreme People's Court Judicial Interpretations:** Hash verification and blockchain archiving can serve as electronic data evidence.
- **Trade Secret Protection:** Encrypted archiving and access control of Layer C satisfy the statutory constituent elements of trade secrets (secrecy, confidentiality, value).

4.4 WIPO International Arbitration System Adaptation

Standardized layered disclosure, chronological evidence preservation, and version traceability structures comply with international IP arbitration evidence submission standards, supporting cross-border originality and first-publication priority burden of proof.

5. Mandatory Standardized Operating Specifications

All scientific achievements publicly published in compliance with IPSTP v1.0 must externally label the following complete elements strictly without exception to ensure evidence chain compliance:

No.	Mandatory Labeling Element	Description
1	Protocol Version (IPSTP v1.0)	Clarify the followed protocol version

2	A/B/C Three-Layer Division Statement	Clarify the scope of public content in Layers A/B; declare Layer C retained as Trade Secret
3	Complete SHA-256 Hash Values of Public Layers A/B	For third-party integrity verification
4	Timestamp Type and Authoritative Source	TSA.cn / Blockchain / DOI, etc.
5	Official Archived DOI	Concept DOI and Version DOI
6	Confirmation Level of this Disclosure	Tier 1 / Tier 2 / Tier 3

Labeling Example (Tier 2 Confirmation):

"This achievement complies with IPSTP v1.0 Tier 2 Evidence Anchoring Level Confirmation. Layer A/B public document SHA-256: [Hash Value]. Zenodo Archived DOI: [DOI]. TSA Timestamp Certificate No.: [Number]. Layer C engineering implementation is retained as a Trade Secret in accordance with the protocol."

6. Protocol Boundaries and Disclaimers

IPSTP does not judge academic rightness or wrongness, directly adjudicate ownership, or automatically determine infringement. Its sole function is to provide a standardized, structured, machine-verifiable, and judicially admissible archiving and disclosure framework.

The ultimate legal validity of all evidence is subject to the official archiving records of the corresponding institutions and the judicial rulings of the competent courts/arbitration institutions.

Appendix A: IPSTP Practical Resources and Implementation Paradigm (Toolchain)

A.1 Preprint and Academic Publication Platforms (Layer A/B Disclosure Infrastructure)

arXiv, OSF Preprints, Research Square, SSRN, Zenodo, Figshare, HAL, bioRxiv, medRxiv, ChemRxiv, EarthArXiv, SocArXiv, engrXiv, TechRxiv, Preprints.org.

A.2 Trusted Temporal Anchoring Tool System (Tiered Chronological Support)

Listed in increasing order of evidence strength:

- **Basic:** DOI official registration timestamp system.
- **Standard:** RFC 3161 standard TSA (TSA.cn), Adobe Trusted Timestamp, National Time Service Center.
- **Enhanced:** OpenTimestamps (Bitcoin anchoring), Ethereum/Polygon blockchain archiving.
- **International:** WIPO Proof international digital evidence service.

A.3 Digital Copyright and Structural Evidence Tool System

- **DOI System:** CrossRef/DataCite dual-DOI issuance.
- **Licensing:** Creative Commons standardized licensing.
- **Version History:** GitHub/GitLab complete version history records.
- **Distributed Storage:** IPFS content-addressed immutable storage, Arweave

perpetual archiving, Filecoin distributed storage.

A.4 Data Hosting and Version Distribution Infrastructure

Public Facilities: GitHub, GitLab, Zenodo-GitHub automatic archiving DOI binding.

Private Preservation: AWS S3 immutable locked storage (WORM mode), local encrypted cold storage (AES-256, offline), off-site multi-backup strategy (at least three independent physical locations).

A.5 Judicial and Arbitration Evidence Toolchain

Third-party electronic data appraisal centers, Notary Office official electronic archiving services, domestic blockchain judicial archiving platforms (Ant Chain, Zhixin Chain, etc.), standardized electronic evidence hash verification tools, cross-border arbitration evidence submission systems.

A.6 Official Recommended Fully Automated VDE Implementation

Paradigm (GitHub+Zenodo Standard Tier 2)

Step 1: Repository Preparation: Create a public GitHub repo; host compliant Layer A/B content; encrypt and package Layer C content into a private directory or independent encrypted storage.

Step 2: Zenodo Association: Log in to Zenodo with GitHub, authorize, and enable the automatic archiving switch for the target repository.

Step 3: Release Publication: Click "Create a new release" in GitHub, fill in the version number (e.g., v1.0.0), upload Layer A/B documents as attachments, and click

"Publish release".

Step 4: Automatic Archiving Trigger: Zenodo automatically captures the snapshot, generates global SHA-256, stores in CERN, records timestamps, and issues dual DOIs (Concept DOI + Version DOI).

Step 5: Verification and Labeling: Obtain DOI and archive hash from Zenodo; fill them into the verification layer of Layer A/B documents; optionally submit the hash to TSA.cn; complete all Tier 2 labeling.

Appendix B: Three-Tier Confirmation Quick Selection Guide

Scenario	Recommended Level	Minimum Operation	Evidence Strength
Rapid First Publication	Tier 1	Preprint + DOI	Academic Archiving Level
Paper Publication	Tier 2	GitHub + Zenodo + TSA	Standard Commercial Level
Core Asset Protection	Tier 3	Multi-anchoring + Notary + Blockchain	Highest Judicial Level
Cross-border Evidence	Tier 3	WIPO Proof + Public Chain Archiving	International Arbitration Level

Protocol License Declaration

The IPSTP v1.0 protocol and this interpretive specification are independently and originally created by Li Zhijun. All researchers can use, modify, and republish them

for free. It is recommended to label the original source for derivative applications.

The protocol does not create new types of legal rights; its evidentiary validity derives from the independent third-party evidence preservation systems it relies on and the existing recognition rules for electronic evidence under current legal frameworks.

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